

People Analytics Case Study: Recruitment Process Optimisation

Hello fellow analyst,

Thank you for progressing to this stage of our interview process. This exercise is designed to give you a sense of the work we do and to assess your analytical and communication skills.

The Scenario

You are an analyst in the People Analytics team. The Head of Talent Acquisition is preparing for a quarterly business review and has asked for your help in understanding the efficiency of our recruitment process over the last year. They are particularly interested in identifying bottlenecks and understanding the return on investment from our different sourcing channels.

Your Task

Using the provided dataset ([applications_data.csv](#)), please prepare a brief analysis that addresses the following questions:

1. **Sourcing Channel Effectiveness:** Which sourcing channels provide the best results? Please evaluate this based on both the volume of hires and the efficiency (conversion rates) of candidates from each source.
2. **Time to Fill:** What is our average time to fill a role (from [date_requisition_opened](#) to [date_hired](#))? Are there significant differences between departments?
3. **Recruitment Funnel Health:** What are the conversion rates between the key stages of our hiring process? The conversion rate for a stage is defined as:
$$\text{Conversion Rate}_X = \frac{\text{Number of candidates entering the previous stage}}{\text{Number of candidates entering Stage X}}$$
Where are the biggest drop-off points?
4. **Recommendations:** Based on your findings, what are your top 2-3 actionable recommendations for the Talent Acquisition team to improve their processes?

Note: At the end of this document, you can find definitions about the datapoints shared.

Deliverable

Please submit a short presentation (maximum 5 slides) or a one-page written summary of your findings. Your submission should include key data visualizations and a clear, concise narrative. The report can take a different format (e.g dashboard) if you prefer.

How do we evaluate the task

We use the same criteria to assess the task that we used to assess our Analysts (you can find more on our analyst career map [here](#)). We'll be looking for:

1. **Actionable Insights:** In Analytics, our goal is to enable better decision-making. We expect your analysis to be practical and centered on the specific actions the product team should take.
2. **Effective Communication:** We value clear communication, and we encourage the use of appropriate data visualisations to effectively convey your findings.
3. **Code Quality:** We appreciate well-structured and readable code. Please ensure your code is easy to follow and understand for those reviewing.

Dataset to work with

You would need a single, granular dataset where each row represents a candidate's application for a specific role.

- **Dataset Name:** `applications_data.csv`
- **Columns:**
 - `application_id`: Unique identifier for each application.
 - `candidate_id`: Unique identifier for each candidate.
 - `job_id`: Unique identifier for the job requisition.
 - `job_title`: e.g., "Data Scientist", "Sales Executive".
 - `department`: e.g., "Technology", "Sales", "Marketing".
 - `job_level`: e.g., "Associate", "Senior", "Manager".
 - `application_date`: Date the candidate applied.
 - `source`: The channel through which the candidate applied (e.g., "LinkedIn", "Employee Referral", "Careers Site", "Agency", "University Job Board").
 - `current_stage`: The final stage the candidate reached in the process (e.g., "Applied", "HR Screen", "Hiring Manager Interview", "Technical Assessment", "Final Interview", "Offer Extended", "Hired", "Withdrew", "Rejected").
 - `date_offer_extended`: Date an offer was made (blank if not applicable).
 - `date_hired`: Date the candidate was marked as hired (blank if not applicable).
 - `date_requisition_opened`: The date the job was first opened.